Industrial Vision

Lecture 09 - Rectification

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Where we are

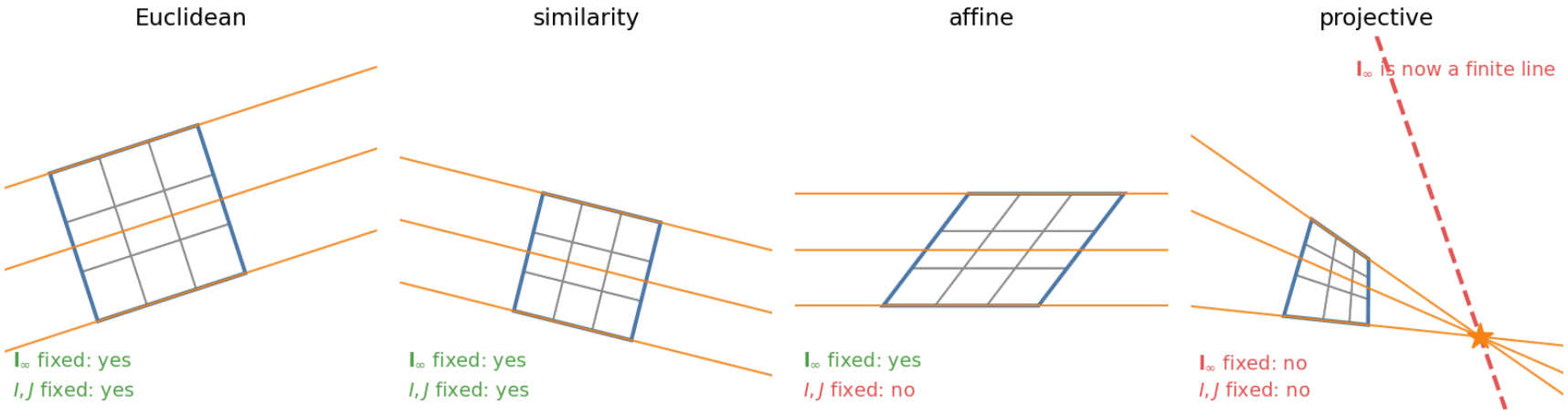
L10 built the projective plane and set out its hierarchy of transformations, from the Euclidean group of 3 degrees of freedom up to the projectivities of 8. That lecture could **describe** the hierarchy. It could not **recover** it.

Consider a photograph of a tiled floor on which every tile is a square. The photograph is a projectivity of that plane, so nothing said so far tells us which lines in the image were parallel on the floor, and nothing tells us which angles were right angles. Every measurement we would like to make has been destroyed, and we have no way to get any of it back.

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The principle: to recover a group, find what it fixes

A transformation group is identified by **what it leaves fixed**. If an entity is fixed under exactly the transformations of a subgroup, then locating that entity in an image tells us how far up the hierarchy we are, and undoing its displacement moves us down.



One square carried through the hierarchy. The orange lines are parallel on the original. Parallelism and the line at infinity survive as far as the affinity; angle and the circular points die one step earlier.

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The line at infinity is fixed under an affinity

Apply L10's line transformation rule $\mathbf{l}' = H^{-\top}\mathbf{l}$ to the affine matrix $H_A = \begin{bmatrix} A & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix}$:

$$\mathbf{l}'_\infty = H_A^{-\top}\mathbf{l}_\infty = \begin{bmatrix} A^{-\top} & \mathbf{0} \\ -\mathbf{t}^\top A^{-\top} & 1 \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \mathbf{l}_\infty$$

The converse also holds. An affinity is the most general linear transformation with $H_{31} = H_{32} = 0$, and that is exactly the condition under which an ideal point is carried to an ideal point.

$$\boxed{\mathbf{l}_\infty \text{ is fixed under } H \iff H \text{ is an affinity}}$$

A projectivity moves it. For $H_P = \begin{bmatrix} A & \mathbf{t} \\ \mathbf{v}^\top & v \end{bmatrix}$ an ideal point does not stay ideal,

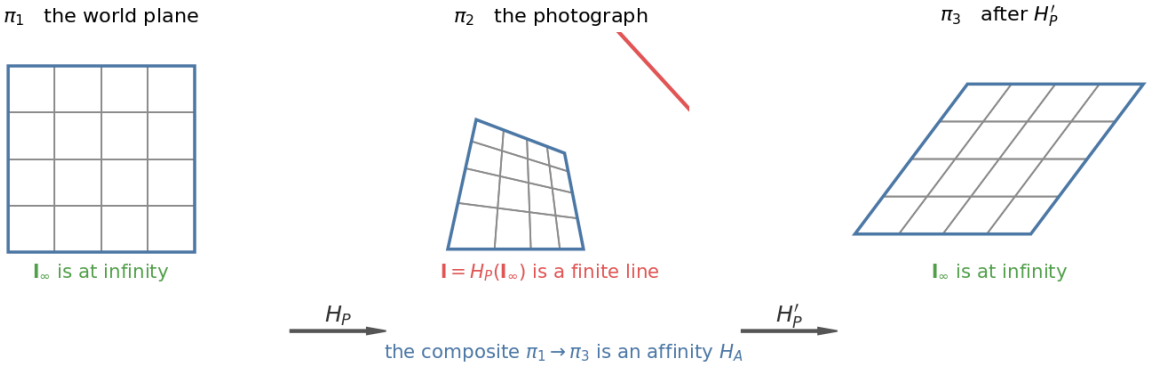
$$H_P \begin{pmatrix} x_1 \\ x_2 \\ 0 \end{pmatrix} = \begin{pmatrix} A\,(x_1, x_2)^\top \\ v_1 x_1 + v_2 x_2 \end{pmatrix},$$

whose third coordinate is in general non-zero. **This is the algebra behind vanishing points.** The horizon in a photograph is the imaged line at infinity of the ground plane, carried to a finite line by the projective part.

Fixed as a set, not pointwise. Under an affinity an ideal point goes to another ideal point, and it is the **same** point only when $A(x_1, x_2)^\top = k(x_1, x_2)^\top$, that is when $(x_1, x_2)^\top$ is an eigenvector of A. The line moves within itself.

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Affine rectification: parallelism from a photograph



Given the imaged line at infinity $\mathbf{l} = (l_1, l_2, l_3)^\top$ with $l_3 \neq 0$ and since $\mathbf{l} = H_p^{-\top}\mathbf{l}_\infty$, we have, $H_p^{\mathrm{T}}[l_1, l_2, l_3]^{\mathrm{T}} = [0, 0, 1]^{\mathrm{T}}$. We can choose,

$$H_p^{\mathrm{T}} = \begin{pmatrix} 1 & 0 & -l_1/l_3 \\ 0 & 1 & -l_2/l_3 \\ 0 & 0 & 1/l_3 \end{pmatrix}.$$

Furthermore, $H_A = H'_p H_p$, so

$$H'_p = H_A H_p^{-1} = H_A \begin{pmatrix} 1 & 0 & -l_1/l_3 \\ 0 & 1 & -l_2/l_3 \\ 0 & 0 & 1/l_3 \end{pmatrix}^{-\mathrm{T}} = H_A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ l_1 & l_2 & l_3 \end{bmatrix}$$

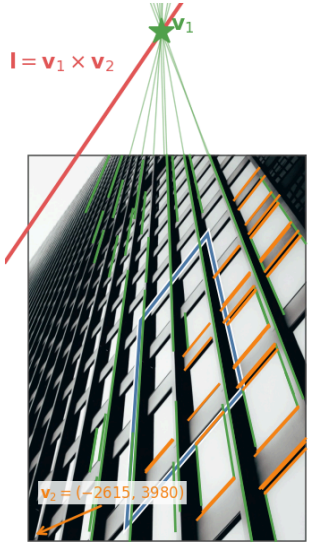
works for **any** affinity H_A, because every affinity already fixes $\mathbf{l}_\infty$.

- Find **two** sets of imaged parallel lines. Each set meets at a point, its **vanishing point**, computed as the null vector of the stacked lines, or by the cross product from two of them.
- The imaged line at infinity is the **join** of those two points, again a cross product.
- Compute H'_P above and **warp** the image by it.
- Parallelism and ratios of lengths along parallel directions are now recoverable.
- **Angles are still wrong**, because the result remains affinely distorted. That is what the rest of the lecture fixes.

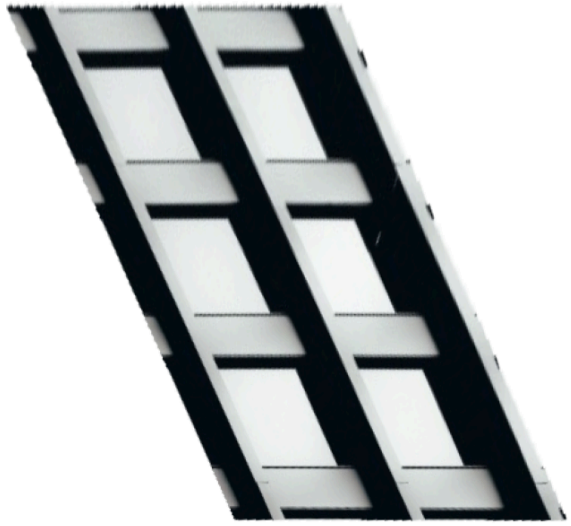
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The same construction, on a real photograph

37 mullions and 46 bands, found by Canny and Hough



the blue bay, affinely rectified

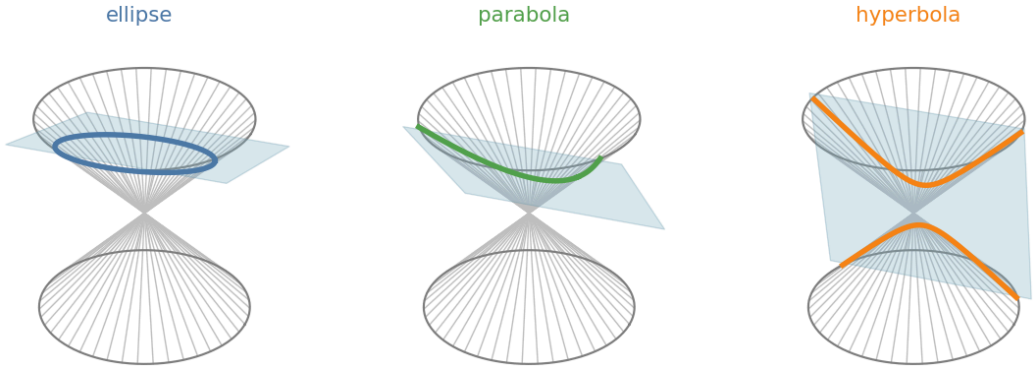


parallel restored, but they meet at 66°, not 90°

- A curtain wall shot from the pavement. The lines were found by Canny and the Hough transform of **L03**, not drawn by hand.
- Their join is the red line, the imaged line at infinity of the wall. The camera was tilted so far up that it **grazes the top left corner of the frame**, and the far edge of the wall runs up to it, which is why the warp is applied to the blue bay and not to the whole picture.
- After the warp both families are parallel and the panels are parallelograms. **They are not rectangles**, because an affinity has been left behind, and that is exactly the ambiguity the next five slides remove.

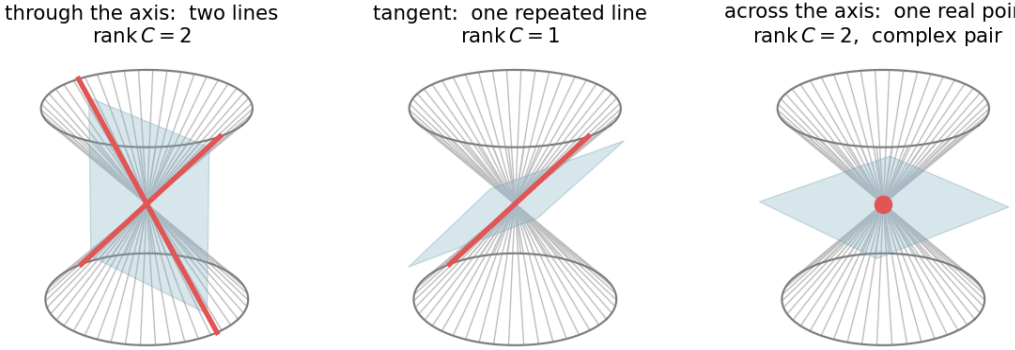
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The cutting plane and the conic



A plane cutting $x^2 + y^2 = z^2$ produces an ellipse, parabola, or hyperbola according to how the plane meets the line at infinity.

Degenerate conics



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Angle is not projectively defined

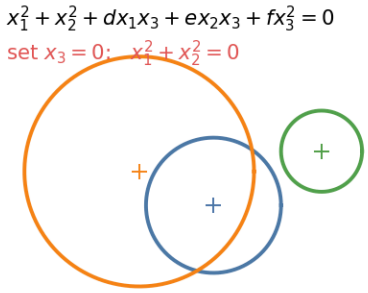
In Euclidean geometry the angle between two lines comes from the inner product of their normals. For $\mathbf{l} = (l_1, l_2, l_3)^\top$ and $\mathbf{m} = (m_1, m_2, m_3)^\top$,

$$\cos\theta = \frac{l_1 m_1 + l_2 m_2}{\sqrt{(l_1^2 + l_2^2)(m_1^2 + m_2^2)}}$$

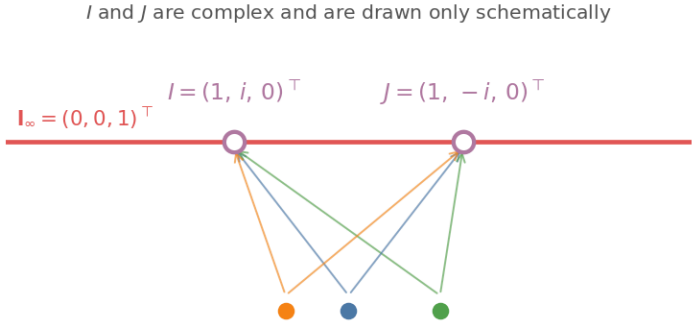
- **This expression is not projectively well defined.** The first two components of a line are not invariant under an affine or a projective transformation of the plane, because the transformed line mixes its third coordinate into its first two. The formula therefore cannot be applied to an image, and the demonstration at the end of the lecture shows it failing on the screen.
- **Example.** The axes $\mathbf{l} = (1, 0, 0)^\top$ and $\mathbf{m} = (0, 1, 0)^\top$ give $\cos\theta = 0$, a right angle. Apply the shear $H_A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. By the line rule $\mathbf{l}' = H_A^{-\top}\mathbf{l} = (1, -1, 0)^\top$, the line $y = x$, while $\mathbf{m}' = (0, 1, 0)^\top$ is unchanged. The same formula now gives $\cos\theta' = -1/\sqrt{2}$, that is 45°. Same two lines, same formula, different number, and by choosing the shear any angle can be produced, so no angle can be assigned to a pair of lines projectively.
- We need a quantity built from $\mathbf{l}$ and $\mathbf{m}$ that is invariant and that reduces to the expression above in a Euclidean frame.

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The circular points I and J



d, *e* and *f* drop out, so the centre and the radius make no difference



every circle, whatever its centre and radius

Under any similarity there are two points on $\mathbf{l}_\infty$ that are fixed. They are the **circular points**, also called the absolute points,

$$\boxed{I = (1,\ i,\ 0)^\top, \qquad J = (1,\ -i,\ 0)^\top}$$

a complex conjugate pair of ideal points.

Why circular. A conic is a circle when $a = c$, scaled to 1, and $b = 0$, so its equation is

$$x_1^2 + x_2^2 + dx_1x_3 + ex_2x_3 + fx_3^2 = 0.$$

Intersect it with $\mathbf{l}_\infty$ by setting $x_3 = 0$:

$$x_1^2 + x_2^2 = 0 \implies (x_1 + ix_2)(x_1 - ix_2) = 0.$$

This is rank-2 conic of two complex lines, met here on $\mathbf{l}_\infty$.

The centre and the radius entered only through d, e and f, and those dropped out. **Every circle meets $\mathbf{l}_\infty$ at the same two points.** They are what all circles have in common and what nothing else has, which is why they carry the metric structure.

Fixed exactly by the similarities. Applying H_S to I returns $se^{-i\theta}I$, which is I up to scale, and likewise for J. A general affinity moves them. So I and J are fixed under H if and only if H is a similarity.

$$H_S = \begin{bmatrix} s\cos\theta & -s\sin\theta & t_x \\ s\sin\theta & s\cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

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- The circular points I, J are fixed points under the projective transformation H if and only if H is a similarity, i.e.

$$I' = H_S I = \begin{bmatrix} s\cos\theta & -s\sin\theta & t_x \\ s\sin\theta & s\cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix}$$

$$= se^{-i\theta} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} = I, \qquad \text{where } e^{i\theta} = \cos\theta - i\sin\theta.$$

- With an analogous proof for J

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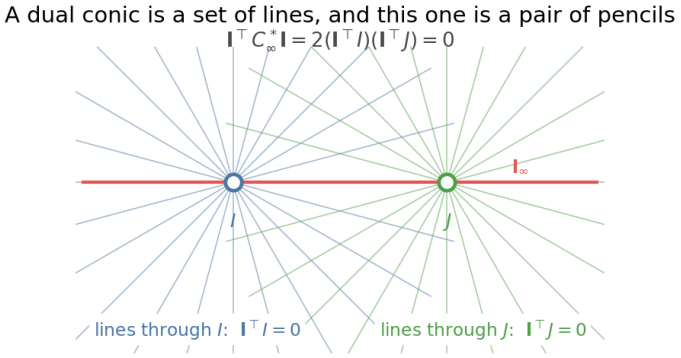
Their dual, the conic C_∞^*

- Working with a pair of complex conjugate points is awkward. Their **dual** is a single **real** matrix, and it is the object that survives into practice.
- **degenerate conics** a rank-2 dual conic *is* a pair of points: it consists of the lines through the one point or through the other.
- A line $\mathbf{l}$ passes through I when $\mathbf{l}^\top I = 0$ and through J when $\mathbf{l}^\top J = 0$, so the object we want is the set of lines with $(\mathbf{l}^\top I)(\mathbf{l}^\top J) = 0$. That product is already quadratic in $\mathbf{l}$, and since a scalar equals its own transpose, $\mathbf{l}^\top I J^\top \mathbf{l} = \mathbf{l}^\top J I^\top \mathbf{l}$:

$$2(\mathbf{l}^\top I)(\mathbf{l}^\top J) = \mathbf{l}^\top \left(IJ^\top + JI^\top\right)\mathbf{l} \implies \boxed{C_\infty^* = IJ^\top + JI^\top}$$

It is **real**, because $IJ^\top + JI^\top = 2\,\mathrm{Re}\,(IJ^\top)$ for a conjugate pair, symmetric by construction, and of **rank 2** as a sum of two rank-one terms. In a Euclidean frame it evaluates to

$$C_\infty^* = (1,i,0)^\top(1,-i,0) + (1,-i,0)^\top(1,i,0) \;\sim\; \mathrm{diag}(1,1,0),$$



The conic itself, drawn as the lines that belong to it. I and J are complex and have no place on the real plane, so they are drawn as two ordinary points of $\mathbf{l}_\infty$.

- **Fixed exactly under a similarity.** A dual conic transforms as $C^{*\prime} = HC^*H^\top$ by L10, and $H_S C_\infty^* H_S^\top = C_\infty^*$ up to scale, with the converse also true.
- $\mathbf{l}_\infty$ **is its null vector.** The circular points lie on $\mathbf{l}_\infty$, so $I^\top \mathbf{l}_\infty = J^\top \mathbf{l}_\infty = 0$ and $C_\infty^* \mathbf{l}_\infty = I(J^\top \mathbf{l}_\infty) + J(I^\top \mathbf{l}_\infty) = \mathbf{0}$.
- C_∞^* **contains** $\mathbf{l}_\infty$ **inside it**, so identifying it does the affine and the metric job at once.

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Orthogonality as $\mathbf{l}'^{\top} C_\infty^{*\prime} \mathbf{m}' = 0$

Once C_∞^* has been identified on the projective plane, Euclidean angles are measured by

$$\boxed{\cos\theta = \frac{\mathbf{l}'^{\top} C_\infty^{*\prime} \mathbf{m}'}{\sqrt{(\mathbf{l}'^{\top} C_\infty^{*\prime} \mathbf{l}')(\mathbf{m}'^{\top} C_\infty^{*} \mathbf{m}')}}}$$

Proof of invariance. Under $\mathbf{x}' = H\mathbf{x}$, we have $\mathbf{l}' = H^{-\top}\mathbf{l}$ and $C_\infty^{*\prime} = H C_\infty^* H^\top$, both from **L10**. The numerator transforms as

$$\mathbf{l}'^{\top} C_\infty^{*\prime} \mathbf{m}' \;\longmapsto\; \mathbf{l}^\top H^{-1}\, H C_\infty^* H^\top\, H^{-\top}\mathbf{m} = \mathbf{l}^\top C_\infty^* \mathbf{m},$$

and the denominator likewise, with the scales of $\mathbf{l}$ and $\mathbf{m}$ cancelling between the two.

In a Euclidean frame $C_\infty^* = \mathrm{diag}(1,1,0)$ and the formula reduces to the expression of the previous slide, as it must. Also,

$$\boxed{\mathbf{l} \text{ and } \mathbf{m} \text{ are orthogonal} \iff \mathbf{l}^\top C_\infty^* \mathbf{m} = 0}$$

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Metric rectification: what the imaged conic knows

Why it works at all. The unknown map from the world plane to the image is a projectivity. Decompose it with the **similarity acting first**, which we are free to do because the Euclidean frame on the world plane is itself only fixed up to a similarity,

$$H = H_P H_A H_S = \begin{bmatrix} I & \mathbf{0} \\ \mathbf{v}^\top & v \end{bmatrix} \begin{bmatrix} K & \mathbf{0} \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} sR & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix}.$$

Then $H_S C_\infty^* H_S^\top = C_\infty^*$ deletes the similarity factor before it is ever used,

$$C_\infty^{*\prime} = H\, C_\infty^*\, H^\top = (H_P H_A)\, C_\infty^*\, (H_P H_A)^\top$$

$$= \begin{bmatrix} I & \mathbf{0} \\ \mathbf{v}^\top & v \end{bmatrix} \begin{bmatrix} K & \mathbf{0} \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} I & \mathbf{0} \\ \mathbf{0}^\top & 0 \end{bmatrix} \begin{bmatrix} K & \mathbf{0} \\ \mathbf{0}^\top & 1 \end{bmatrix}^\top \begin{bmatrix} I & \mathbf{0} \\ \mathbf{v}^\top & v \end{bmatrix}^\top$$

$$= \begin{bmatrix} KK^\top & KK^\top \mathbf{v} \\ \mathbf{v}^\top KK^\top & \mathbf{v}^\top KK^\top \mathbf{v} \end{bmatrix}$$

The image of C_∞^* carries the projective part $\mathbf{v}$ and the affine part K, and **nothing of the similarity part**. Identifying C_∞^* in an image therefore determines the distortion exactly up to the similarity.

From the conic to the homography. The matrix above is **not** a homography but an object we can *measure*, because each pair of world-orthogonal imaged lines gives one linear constraint $\mathbf{l}^\top C_\infty^{*\prime} \mathbf{m} = 0$ on its six entries. The rectifying homography is obtained from it by one demand: **a rectified image is one in which the imaged conic sits back in canonical position.**

$$\text{find } H_r \quad \text{with} \quad H_r\, C_\infty^{*\prime}\, H_r^\top = \operatorname{diag}(1,1,0) = C_\infty^*$$

- Since $C_\infty^{*\prime} = H C_\infty^* H^\top$, factoring the measured matrix as $C_\infty^{*\prime} = A \operatorname{diag}(1,1,0)\, A^\top$ recovers the distorting map, $A = H$, and $H_r = A^{-1}$ undoes it. The factorisation is a real linear-algebra operation on a real symmetric rank-2 matrix, done in two ways.
- **It is solvable only up to a similarity.** If H_r works then so does $H_S H_r$, because H_S is *defined* by fixing $\operatorname{diag}(1,1,0)$. That non-uniqueness is not slack in the method; it is the residual ambiguity of the metric stratum, reported honestly.

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Two routes to H_r

Two steps. On an affinely rectified image only H_A remains, so $\mathbf{v} = \mathbf{0}$ and the measured conic is $C_\infty^{*\prime} = \begin{bmatrix} S & \mathbf{0} \\ \mathbf{0}^\top & 0 \end{bmatrix}$ with $S = KK^\top$. For two lines $\mathbf{l}, \mathbf{m}$ **measured in that affinely rectified image**, the orthogonality test $\mathbf{l}^\top C_\infty^{*\prime} \mathbf{m} = 0$ loses its third coordinate and reads

$$\begin{pmatrix} l_1 & l_2 \end{pmatrix} S \begin{pmatrix} m_1 \\ m_2 \end{pmatrix} = 0.$$

S is symmetric with three entries and hence **2 degrees of freedom** after scale, so **two** orthogonal pairs determine it as a null vector.

The factorisation is then immediate: Cholesky gives $S = KK^\top$, so $A = \begin{bmatrix} K & \mathbf{0} \\ \mathbf{0}^\top & 1 \end{bmatrix}$ and

$$H_r = \begin{bmatrix} K^{-1} & \mathbf{0} \\ \mathbf{0}^\top & 1 \end{bmatrix}, \qquad H_r C_\infty^{*\prime} H_r^\top = \begin{bmatrix} K^{-1} K K^\top K^{-\top} & \mathbf{0} \\ \mathbf{0}^\top & 0 \end{bmatrix} = \operatorname{diag}(1, 1, 0$$

One step. Nothing above required the image to be rectified first. Applied to the perspective image, each orthogonal pair $\mathbf{l}, \mathbf{m}$ **measured there** gives, through the same test $\mathbf{l}^\top C_\infty^{*\prime} \mathbf{m} = 0$, one linear constraint on the six entries of $C_\infty^{*\prime}$,

$$\left(l_1 m_1,\ \tfrac{l_1 m_2 + l_2 m_1}{2},\ l_2 m_2,\ \tfrac{l_1 m_3 + l_3 m_1}{2},\ \tfrac{l_2 m_3 + l_3 m_2}{2},\ l_3 m_3\right) \mathbf{c} = 0,$$

so **five** pairs give a 5×6 matrix whose null vector is $C_\infty^{*\prime}$.

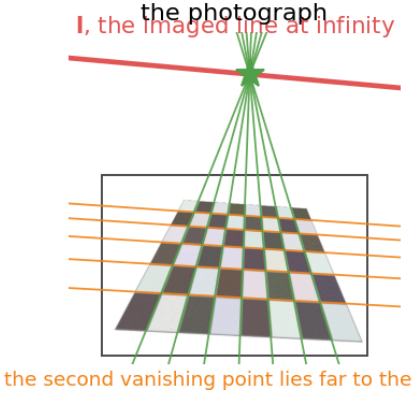
It comes back real, symmetric and of rank 2, so its eigendecomposition $C_\infty^{*\prime} = U \operatorname{diag}(\sigma_1, \sigma_2, 0)\, U^\top$ *is* the factorisation we asked for,

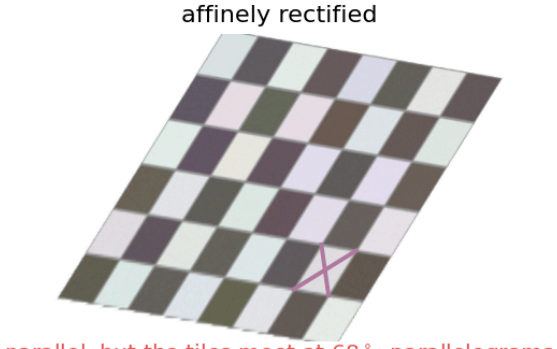
$$A = U \operatorname{diag}(\sqrt{\sigma_1}, \sqrt{\sigma_2}, 1), \qquad H_r = A^{-1} = \operatorname{diag}\left(\sigma_1^{-1/2}, \sigma_2^{-1/2}, 1\right) U^\top,$$

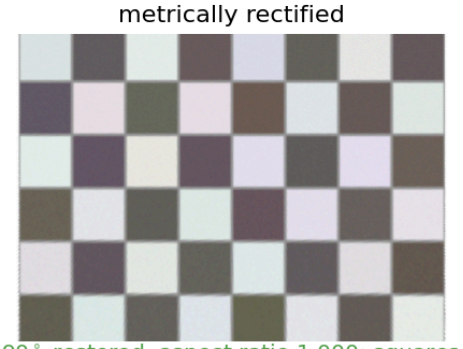
since $A \operatorname{diag}(1, 1, 0)\, A^\top = U \operatorname{diag}(\sigma_1, \sigma_2, 0)\, U^\top$. Both distortions come out together. The 1 in the third slot is arbitrary, and a different choice changes H_r by a scaling, which is a similarity.

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Both rectifications, on one photograph







- The tiles are square on the floor, so their **diagonals** are orthogonal too. That second pair is what fixes the aspect ratio, and it is why this example is a floor rather than the facade, which offers only one pair of orthogonal directions.
- Applied directly to the left image, five orthogonal pairs returned $C_\infty^{*\prime}$ with singular values $(1,\ 0.349,\ 0.005)$. The rank-2 condition was never imposed and appeared on its own.

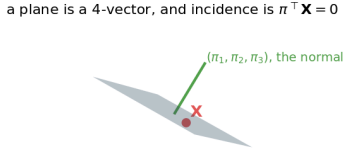
Up one dimension: $\mathbb{P}^3$, points, planes and duality

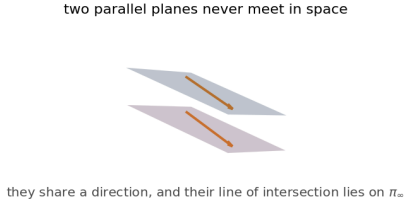
A point of $\mathbb{P}^3$ is a homogeneous 4-vector $\mathbf{X} = (X_1, X_2, X_3, X_4)^\top$, representing the Euclidean point $(X_1/X_4,\ X_2/X_4,\ X_3/X_4)$ when $X_4 \neq 0$ and an **ideal point**, that is a direction, when $X_4 = 0$.

A plane is also a homogeneous 4-vector $\boldsymbol{\pi} = (\pi_1, \pi_2, \pi_3, \pi_4)^\top$, whose first three entries are the Euclidean normal, and incidence is the same scalar product as in $\mathbb{P}^2$ with one coordinate added,

$$\boxed{\boldsymbol{\pi}^\top \mathbf{X} = 0}$$

In $\mathbb{P}^3$ the dual of a point is a plane, not a line. The natural guess is wrong and it will be made. A line in $\mathbb{P}^3$ is self-dual and awkward to represent.





A projective transformation of $\mathbb{P}^3$ is a non-singular 4×4 matrix with **15 degrees of freedom**, and points and planes transform by

$$\mathbf{X}' = H\mathbf{X}, \qquad \boldsymbol{\pi}' = H^{-\top}\boldsymbol{\pi},$$

by the argument of **L10** with one coordinate added.

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The plane at infinity π_∞

$$\boxed{\boldsymbol{\pi}_\infty = (0,0,0,1)^\top}$$

It contains exactly the directions $\mathbf{D} = (X_1, X_2, X_3, 0)^\top$, and it is the analogue of $\mathbf{l}_\infty$ in every respect.

$\boldsymbol{\pi}_\infty$ is a fixed plane under H if and only if H is an affinity. The proof is the analogue to before. As before it is fixed **as a set and not pointwise**, and under a particular affinity other planes may also be fixed; only $\boldsymbol{\pi}_\infty$ is fixed under **every** affinity.

- Two planes are parallel if and only if their **line of intersection lies on $\boldsymbol{\pi}_\infty$**.
- A camera images $\boldsymbol{\pi}_\infty$ as the **horizon**, and an ideal point of a world plane as a **vanishing point**.

Locating $\boldsymbol{\pi}_\infty$ in a projective reconstruction, and moving it to $(0,0,0,1)^\top$, upgrades the reconstruction to affine.

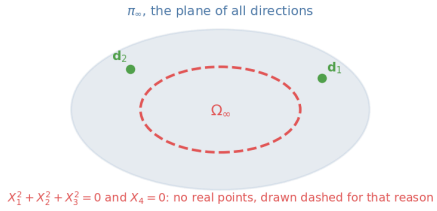
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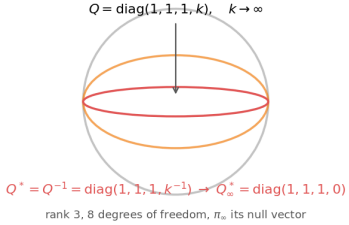
The absolute conic Ω_∞ and its dual Q^*_∞

The absolute conic is a **point conic lying on $\boldsymbol{\pi}_\infty$**. In a metric frame it needs **two** equations,

$$\boxed{X_1^2 + X_2^2 + X_3^2 = 0 \quad \text{and} \quad X_4 = 0}$$

- a conic in space is a curve, so it requires both a surface and the plane it lies in. Restricted to $\boldsymbol{\pi}_\infty$ it is the matrix $I_{3\times 3}$, so the condition on a direction is $\mathbf{d}^\top \mathbf{d} = 0$ and Ω_∞ **has no real points**.
- Ω_∞ is fixed under H if and only if H is a similarity.
- A transformation fixing Ω_∞ must fix its support plane, so it is affine. Restricted to $\boldsymbol{\pi}_\infty$ the conic is I, and by L10's conic rule being fixed means $A^{-\top} I A^{-1} = I$ up to scale. Taking inverses gives $AA^\top = I$, so A is orthogonal, a scaled rotation with or without a reflection.





Angles between directions are measured exactly as on slide 9, one dimension up,

$$\cos\theta = \frac{\mathbf{d}_1^\top \Omega_\infty \mathbf{d}_2}{\sqrt{(\mathbf{d}_1^\top \Omega_\infty \mathbf{d}_1)(\mathbf{d}_2^\top \Omega_\infty \mathbf{d}_2)}},$$

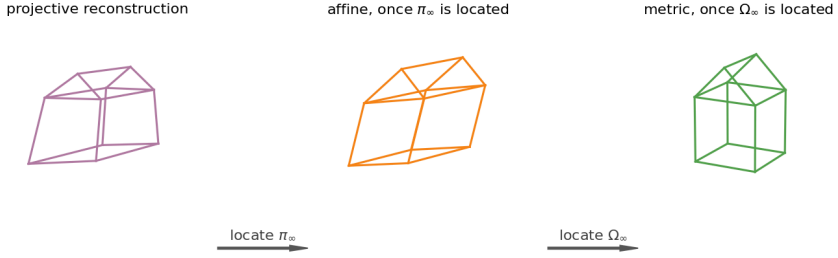
and this becomes **L14's** calibration constraint once Ω_∞ is replaced by its image ω under KR.

The dual. Q^*_∞ is the set of planes tangent to Ω_∞: rank 3, **8 degrees of freedom**, being ten entries less one for scale and one for $\det = 0$. It **packages $\boldsymbol{\pi}_\infty$ and Ω_∞ into one object**, since $\boldsymbol{\pi}_\infty$ is its null vector, so one matrix recovers both strata.

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The three strata

Stratum	Fixed entity, plane and space	dof, plane / space	What is restored
Projective	none	8 / 15	straightness, concurrency, the cross ratio
Affine	$\mathbf{l}_\infty \cdot \boldsymbol{\pi}_\infty$	6 / 12	parallelism, ratios of lengths along parallels
Metric	$C^*_\infty \cdot \Omega_\infty, Q^*_\infty$	4 / 7	angle, ratios of lengths

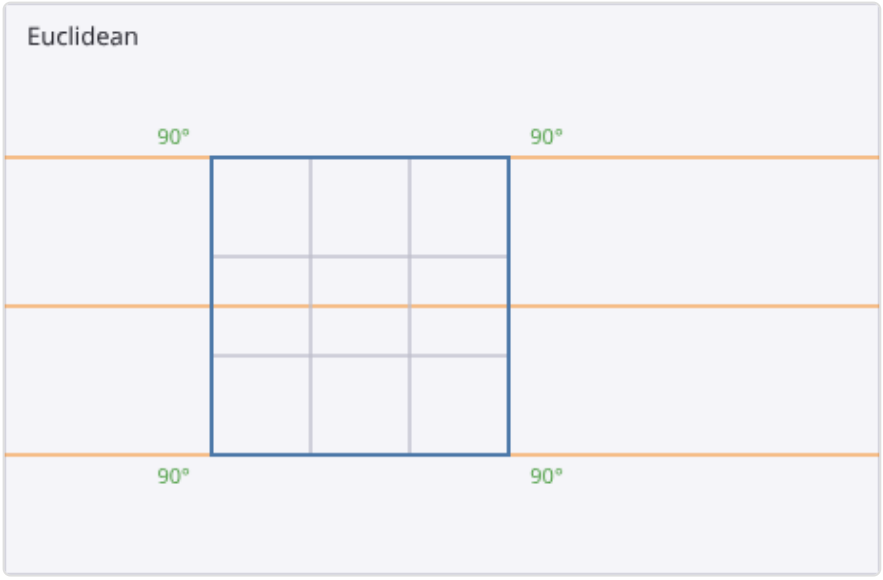


- The ladder is climbed downwards by locating entities, each removing exactly the freedom that separates one stratum from the next and no more.
- **L14** finds the image of Ω_∞ by calibrating a camera, and calls it ω.
- **L20** uses $\boldsymbol{\pi}_\infty$ to reach affine and then ω to reach metric. The house on the left is that lecture in three pictures.

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Inline demonstration

what is fixed | rectification



sweep

l∞ fixed	yes
I, J fixed	yes
corner error from 90°	0.0°
opposite corners differ by	0.0°

The blue square and its grid are one figure carried through the hierarchy, and the four numbers on it are its interior angles. The orange lines are parallel on the original.

The two rows above separate the two entities exactly. **Angle** dies at the affinity, which is the moment **I** and **J** stop being fixed. **Opposite angles keep agreeing** through the affinity, because an affinity carries a square to a parallelogram, and part company only at the projectivity, which is the moment $\mathbf{l}_\infty$ stops being fixed and becomes the dashed red line through the vanishing point.

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Questions

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Summary

- **To recover a group, find the entity fixed under it.** One principle, five instances, and the whole lecture.
- $\mathbf{l}_\infty$ is fixed if and only if H is an affinity, and fixed **as a set** rather than pointwise. A projectivity carries it to a finite line, which is why photographs have horizons.
- **Affine rectification.** Two sets of imaged parallel lines give two vanishing points; their join is the imaged $\mathbf{l}$, and the matrix with $\mathbf{l}^\top$ as its last row sends it back to $(0,0,1)^\top$. Angles remain wrong and a free affinity remains.
- The Euclidean angle formula is **not projectively defined**. Its replacement is $\cos\theta = \mathbf{l}^\top C_\infty^* \mathbf{m} / \sqrt{(\mathbf{l}^\top C_\infty^* \mathbf{l})(\mathbf{m}^\top C_\infty^* \mathbf{m})}$, and orthogonality is $\mathbf{l}^\top C_\infty^* \mathbf{m} = 0$.
- The **circular points** $I = (1, i, 0)^\top$ and $J = (1, -i, 0)^\top$ are where every circle meets $\mathbf{l}_\infty$, and are fixed exactly by the similarities.
- Their dual $C_\infty^* = IJ^\top + JI^\top$ is a real rank-2 line conic with **4 degrees of freedom** and $\mathbf{l}_\infty$ as its **null vector**, so it carries the affine information as well as the metric.
- **Metric rectification.** The image of C_∞^* carries $\mathbf{v}$ and K but nothing of the similarity, so the rectifying homography is any H_r with $H_r C_\infty^{*\prime} H_r^\top = \operatorname{diag}(1,1,0)$, fixed only up to that similarity. Measure the conic, factor it as $A \operatorname{diag}(1,1,0)\, A^\top$, warp by A^{-1}: two orthogonal pairs suffice after affine rectification, five suffice directly.
- In $\mathbb{P}^3$ points and planes are both 4-vectors and **dual**; $\boldsymbol{\pi}_\infty = (0,0,0,1)^\top$ is fixed exactly by the affinities; Ω_∞, needing two equations, and $Q_\infty^* = \operatorname{diag}(1,1,1,0)$ of rank 3 and 8 dof, exactly by the similarities.
- **The strata:** projective, then affine once $\mathbf{l}_\infty$ or $\boldsymbol{\pi}_\infty$ is known, then metric once C_∞^* or Ω_∞ is known.

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References

1. R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge Univ. Press, 2004. See §2.7 on the line at infinity, the circular points, C_∞^*, angles and both rectifications, and Ch. 3, §3.1–§3.7.
2. Liebowitz, David, and Andrew Zisserman. "Metric rectification for perspective images of planes." Proceedings. 1998 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (Cat. No. 98CB36231). IEEE, 1998.

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